VOLUME Theorems

Theorems of the AB Research Program

Th1–Th6: hermiticity, self-adjointness, certificates, the spectral Li formula, block reduction, the Guinand–Weil trace formula

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Th1. Hermiticity and reality of the spectrum

Theorem Th1 (V4).

For any choice of vortices, W, α and boundaries the operator H_L satisfies H_L = H_L†; hence spec(H_L) ⊂ ℝ. At α = ½ and W = 0 the spectrum is symmetric: λ ∈ spec ⇒ −λ ∈ spec.

> ***Proof.*** *Every bond is inserted as the pair H[i,j] = −t·e^{iφ}, H[j,i] = −t·e^{−iφ} = conj(H[i,j]) — hermiticity by construction; the diagonal is real (vortex potential, disorder, staggered mass). By the spectral theorem for Hermitian matrices the whole spectrum is real. Symmetry at W = 0: the lattice is bipartite in the parity ix+iy; in that basis H = [[0, B], [B†, 0]]; if H(λ, 0) = λ·(u, v) then H(−v, u)·(−1) = −λ·(u, v) — the spectrum is symmetric about zero. Numerical certificate: ‖H − H†‖ = 0.0 exactly; the pairing λ ↔ −λ reproduced to 8.2e-15; reality residuals 8.8e-15 (the eigvalsh machine floor).*

Th1.1 Certificate and boundaries

The certificate is reproducible: test_th1_passport.py checks hermiticity on three protocol gates (Nv = 16/64/256, seed 96) and demands ‖H − H†‖ ≤ 1e-12; the fact is exactly 0.0, since the construction uses no approximations — conjugate pairs are written simultaneously. Importantly, hermiticity survives ALL the C5 handles — twists, anisotropy, δα, staggered mass, profile, signs: each handle either modifies the bond magnitude (preserving the phase pair) or the diagonal (staying real). This makes Th1 an architectural property, not a property of a single configuration.

Boundaries: Th1 says nothing about the correspondence of the spectrum to the ζ zeros — that is the H2/H5 question, on which the series delivers negative verdicts (V1, V3, V6, V16). The pairing λ ↔ −λ breaks already at W > 0 (the vortex potential is asymmetric) and is not restored by vortices.

Th2. Finite-dimensional self-adjointness

Theorem Th2 (V8).

A symmetric operator on ℂ^N is essentially self-adjoint: defect indices (0, 0); e^{−iHt} is unitary; Im⟨ψ, Hψ⟩ = 0.

> ***Proof.*** *On ℂ^N every bounded symmetric operator is self-adjoint: D(H) = ℂ^N and the defect spaces ker(H* − iI), ker(H* + iI) are empty, since ‖(H ∓ iI)ψ‖² = ‖Hψ‖² + ‖ψ‖² ≥ ‖ψ‖² > 0 (the cross-term vanishes by symmetry). Consequences: real spectrum, e^{−iHt} a unitary group, ⟨ψ, Hψ⟩ real. Numerically: the minimal singular value of H − iI is 3.9 (theory: √(1 + min‖Hψ‖²) ≥ 1), the unitarity drift of e^{−iHt} is 2.2e-16 over 1000 steps. The limit part (L → ∞, essential self-adjointness of an unbounded operator) is the open obligation O03.*

Th2.1 Certificate and boundaries

Certificate: test_th2_selfadjoint.py checks three equivalent manifestations of self-adjointness — empty defects, unitary dynamics, real form — on the protocol gate; all three hold at machine floors. Practical value: the unitarity of e^{−iHt} underpins the dynamics campaigns (V17): the energy conservation 1.8e-15 in V17 is a direct corollary of Th2. The gap "finite-dim proved / limit open" is recorded honestly: in the

infinite-dimensional problem symmetry is not enough, boundary conditions are needed (open boundaries are natural, but the strict convergence of resonances is an open problem of the program).

Boundaries: the theorem is finite-dimensional; the transfer to L → ∞ requires estimates of the essential spectrum and resonance geometry — see O03 in the obligation DAG.

Th3. Eigenvalue certificates

Theorem Th3 (V12).

For Hermitian H, a computed $\tilde\lambda$, a unit v with residual r = Hv − $\tilde\lambda$v, δ = ‖r‖ and gap g to the nearest other level, some $\hat\lambda \in$ spec H satisfies $|\hat\lambda - \tilde\lambda| \le \delta^2/g$; the whole spectrum lies in the union of rational Gershgorin intervals.

Proof. *Decompose v = c·u + w with u the normalized eigenvector of $\hat\lambda$ and w ⊥ u. The residual weight on foreign levels yields the Kato-type estimate ‖w‖ ≤ δ/gap, hence $|\hat\lambda - \tilde\lambda| \le \delta^2/gap$ — quadratic superconvergence: at δ ~ 1e-8 and gap ~ 0.5 the interval shrinks to ~2e-16. Gershgorin: for H = D + E the radii R_i = Σ_{j≠i}\|H_ij\| in exact rational arithmetic (fractions, limit_denominator 1e12) give rounding-free intervals; their union covers the spectrum deterministically. Verified 5/5 certificates with width ~1e-27 in mpmath (dps 40).*

Th3.1 Certificate and boundaries

Certificate: test_th3_certificates.py checks (a) the claimed level falls into its Kato interval, (b) the spectrum is covered by Gershgorin intervals, (c) the width reproduces. The series threshold 1e-27 is achieved on the first five levels of the protocol gate. Value for the program: the interval layer is a "trust passport" for all spectral comparisons (V1–V3, V6, V16) — any coincidence of levels with zero ordinates can be certified by intervals, not only by floating-point numbers.

Boundaries: the estimate is quadratic in δ and therefore requires an already well-converged $\tilde\lambda$; for degenerate or close levels the gap is small and the interval widens — cluster versions of the estimate are not included in the package. Rational arithmetic is slow: the layer is applied pointwise, not to the whole spectrum.

Th4. The spectral Li formula

Theorem Th4 (V15).

The Li coefficients admit the spectral representation λ_n = Tr g_n(H_ξ), g_n(E) = 1 − (1 − 1/ρ(E))^n, ρ(E) = ½ − iE/2; the series converges absolutely; two computational paths agree with |Δ| ≤ 4.4e-11 for n ≤ 3000.

Proof. *By definition λ_n = Σ_ρ [1 − (1 − 1/ρ)^n] over the nontrivial zeros with conjugation accounted (λ_n is real). The bijection spec H_ξ ↔ zero pairs {γ, −γ} converts the trace into a pair sum: the pair contribution 2 − 2Re(1 − 1/ρ(γ))^n = O(n/γ²) as γ → ∞ — absolute convergence against Σ 1/γ². The poles of g_n sit at ρ = 1, i.e. E = −i — off the real spectrum of H_ξ: the trace is well defined. Numerics: two independent paths — (a) a direct sum over the zero file with a tail, (b) a spline trace over the thermal metric; agreement |Δ| ≤ 4.4e-11. Positivity λ_n ≥ 0 — the Li criterion — is measured for n ≤ 3000 (tails ≤ 0.115); for all n the question is open and equivalent to RH.*

Th4.1 Certificate and boundaries

Certificate: test_th4_li.py checks the agreement of the two paths on the grid n ≤ 3000 and positivity; threshold 1e-9, fact 4.4e-11. Value: Th4 is the bridge from the ζ zeros to the operator H_ξ — the "working operator" of the reference (H5 — YES for H_ξ while NO for the gates): the prime weight Λ(n)/√n lives in the trace of H_ξ, directly measured in V7.

Boundaries: absolute convergence is proved, but positivity is empirical only up to n = 3000; extending the grid is linear in time and bounded by the zero file (external data). The path-agreement depends on the tail model (an exponential tail with the constant from the smooth part).

Th5. Block reduction by symmetry

Theorem Th5 (V9).

If [H, A] = [H, B] = 0 for the generators of the abelian group G of magnetic translations, then H is block-diagonal in the character decomposition of G and spec H = ⊔_χ spec(H|Π_χ ℂ^N).

> ***Proof.*** *Commuting unitaries A, B are simultaneously diagonalizable; their joint eigenvalues are the characters χ: G → U(1); the spectral projectors Π_χ (polynomials in A, B) commute with H: [H, Π_χ] = 0, hence each Π_χ ℂ^N is invariant and H decomposes into the direct sum of restrictions. Corollary: eigenvalues are found in smaller blocks with guaranteed coverage. Numerically: the union of 6 blocks reproduces the full spectrum to 7.1e-15; vortices break the symmetry (commutators ε_g = O(1)) — the reduction applies only to the clean lattice and to symmetric vortex configurations.*

Th5.1 Certificate and boundaries

Certificate: test_th5_block_reduction.py compares the union of the block spectra with the full spectrum; threshold 1e-9, fact 7.1e-15. The value is twofold: (a) computational — the reduction speeds up scans (each block ~$L^2/6$ instead of L^2); (b) diagnostic — the magnitude ‖[H, A]‖ measures how far a vortex configuration is from symmetric and serves as a passport feature of a gate in V4.

Boundaries: for protocol (asymmetric) vortices the reduction does not apply — this is not a defect but the boundary of its domain; partially symmetric configurations give approximate invariant subspaces, and the quantification of this idea is not included in the package.

Th6. The Guinand–Weil trace formula

Theorem Th6 (V7).

$Q(a) = \Sigma_{\gamma>0}\, 2a/(a^2+\gamma^2) = \zeta'/\zeta(\tfrac{1}{2}+a) + 1/(\tfrac{1}{2}+a) + 1/(a-\tfrac{1}{2}) - \tfrac{1}{2}\ln\pi + \tfrac{1}{2}\psi(\tfrac{1}{4}+a/2) - B - C_0$, with $B = \tfrac{1}{2}\ln(4\pi) - 1 - \gamma/2$, $C_0 = -B$; and $K(x) = \mathrm{Tr}\, e^{-xH_\Xi} = (1/\pi)\int_0^\infty Q(a)\cdot\sin(ax)\, da$.

Proof. *Step 1: the logarithmic derivative of the Hadamard product $\xi(s) = \tfrac{1}{2}\cdot\prod_\rho(1 - s/\rho)$ gives $\xi'/\xi(\tfrac{1}{2}+a)$ proportional to $\Sigma_\gamma\, 2a/(a^2+\gamma^2)$ — the sum $Q(a)$ up to the normalization of the poles. Step 2: the explicit pole terms (the ζ pole at $s = 1$, the functional equation at $s = \tfrac{1}{2}$, the Γ-factor through ψ) assemble into the constants; the comparison at $a \to 0+$ fixes $B = \tfrac{1}{2}\ln(4\pi) - 1 - \gamma/2$ (ERRATUM: the parent repository encodes $\tfrac{1}{2}\ln(2\pi)$; here the value is re-derived and checked by the asymptotics). Step 3: the representation $2a/(a^2+\gamma^2) = 2\int_0^\infty e^{-\gamma t}\cdot\sin(at)\, dt$ and the exchange of integration (absolute convergence at $x > 0$) give $K(x) = (1/\pi)\int Q(a)\, \sin(ax)\, da$. Step 4: at $a > \tfrac{1}{2}$ the series $\zeta'/\zeta = -\Sigma\, \Lambda(n)/n^{\tfrac{1}{2}+a}$ transfers the prime weight into the trace: the prime part of K is the comb over $\Lambda(n)/\sqrt{n}$. Numerically: $|\Delta Q| \le 4.6\text{e-}8$ (mpmath, 601-node spline), $|\Delta K| \le 6.9\text{e-}6$ (Λ-sieve to $2\cdot10^6$, tail $\le 1\text{e-}6$).*

Th6.1 Certificate and boundaries

Certificate: test_th6_trace_formula.py checks Q(a) against the direct summation over the zero file and K(x) against the prime part; thresholds 1e-6/1e-5, facts 4.6e-8/6.9e-6; the PC0 recipe $u\cdot\Sigma e^{-u\mathcal{N}(\gamma)} - (1 - e^{-uX})$ reproduced with the correlation r = +0.9996. Value: Th6 is the most precise bridge of the series "zeros ↔ operator": it not only records THAT primes live in the trace of H_Ξ but also delivers the numerical pipeline (V7) on which the null operator of the program is assembled.

Boundaries: the constant B depends on the normalization of Ξ; the formula for K(x) requires $x > 0$ and the smooth part with exact Si/Ci forms; the Λ-sieve to $2\cdot10^6$ limits the comb resolution — the tail beyond $n > 2\cdot10^6$ is estimated by an exponential, its contribution ≤ 1e-6.

The status table

Table 1. Theorem statuses

Theorem	Campaign	Status	Key number
Th1	V4	proved	$\|H-H^\dagger\| = 0$
Th2	V8	proved (finite-dim)	drift 2.2e-16
Th3	V12	proved + verified	5/5 certificates
Th4	V15	proved + verified	$\|\Delta\| = 4.4\text{e-}11$
Th5	V9	proved + verified	7.1e-15
Th6	V7	proved + verified	$\|\Delta K\| = 6.9\text{e-}6$

The boundary of the proved

None of the theorems proves RH or establishes the spectral identity for the lattice gates — the latter question is measured negatively three times (V1, V6, V16 — over the entire full handle space). The exact status of the program is fixed by the obligation DAG (V13): the path to RH contains 4 conjectures; the open obligations O03 (the $L \to \infty$ limit), O18 (CNOT synthesis on the blocks), O19 (embedding the pulses into the full lattice) are added in v1.1.0. The theorems of the package are the foundation and the measurement infrastructure, not a final result.

Six theorems, each with a full derivation, a numerical certificate, a reproducible test and honest boundaries. This is the honesty discipline in action.